

Early Breast Cancer Screening from Thermography via Deep Pre-Trained Edge Detection with Extreme Gradient Boosting

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Abstract—Breast cancer is the most common type of cancer affecting women worldwide. Early diagnosis is crucial for successful treatment, recovery, and mortality reduction. Therefore, various non-invasive, minimally invasive, and invasive approaches have been developed for breast cancer detection. Breast thermography is a promising non-invasive, low-cost, and radiation-free imaging technique that can provide information about the physiological variations of breast tissue. It can be considered as a supplement to X-ray mammography or even a primary early cancer detection tool, particularly for women with dense breast tissue or those who are not suitable candidates for mammography. This study proposes a fully automated system for thermography-based early breast cancer detection. The characteristic features of the breast are extracted based on the deep pre-trained holistically-nested edge detector. Such features are used to diagnostically classify breast tissue into normal or malignant classes using the machine learning algorithm extreme gradient boosting. The classification improved to some extent by incorporating the Canny edge detector. The developed system achieved an accuracy of 97.4%, with a precision of 95%, a recall of 100%, and an F1 score of 97%.

Index Terms—Breast cancer; thermography; Holistically-nested edge detector; extreme gradient boosting.

I. INTRODUCTION

Breast cancer (BC) is a consequence of abnormal behavior in the functionality of normal breast cells. Disturbance in the ordinary properties of breast cells can occur due to various reasons, including genetic mutations, hormonal changes, and environmental factors. These changes can cause the breast cells to grow and divide uncontrollably, leading to the development of a mass or lump in the breast, forming either a benign or a malignant tumor [1]. Unfortunately, BC is the second most prevalent cancer worldwide and is considered one of the most popular and frequent types of cancer among women. Besides, it is the leading cause of women's cancer deaths [2]. According to 2022 cancer statistics adopted from the American Cancer Society, BC incidence rate increased by 0.5% per year. The average 5-year and 10-year survival rates for women with breast cancer are 90% and 84%, respectively. In contrast, the 5-year survival rate decreases to 85% if the BC spreads to the regional lymph nodes and 26% if it extends to a distant body part. In addition, the estimated new cases diagnosed for breast cancer and deaths in women will be 31%

and 15% by the end of 2021 [3]. Therefore, early detection of BC has become an important research topic due to the remarkable change in mortality rates. Besides, improving the quality of life and increasing the number of available treatment options is crucial.

Currently, many imaging modalities have been used for BC screening. Among them, breast thermography is non-invasive, low-cost, and radiation-free. It uses an infrared camera to capture the inherent body's infrared radiation, which varies according to the physiological changes in breast tissue. The use of thermography for BC identification was first suggested by Lawson et al. [4]. It was noticed that the local temperatures with the area of tumor masses were much higher than those of normal skin. Therefore, it can be considered as a supplement to X-ray mammography or even a primary early cancer detection tool, particularly for women with dense breast tissue or those who are not suitable candidates for mammography.

Improving the ability of thermography imaging for early BC diagnosis and increasing the chances of successful treatment is toward computer-aided diagnosis. Morais et al. [5] proposed a conjugated gradients method to identify tumor lesions in breast tissue through thermography. Using the approach, 96% of the 47 breast abnormalities that were evaluated were identified. In [6], the authors could precisely segment the two breasts for asymmetry analysis by locating the breast borders on the thermal images via a projection profile approach. By employing morphological watershed after automatic segmentation, the warmest areas in breast thermal images were highlighted [7]. Such an approach achieved segmentation accuracy, sensitivity, and specificity of 90.3%, 86%, and 89%, respectively. Marcus et al. [8] combined morphological image processing and symbolic data analysis model for interval feature extraction from breast thermal images. The features were then transformed by Fisher's criterion for the classification process of breast lesions. The model misclassification rate, sensitivity, and specificity were 16%, 85.7%, and 86.5%, respectively. Moreover, Resmini et al. [9] utilized genetic algorithms along with supervised and unsupervised learning approaches for BC detection.

This study proposes a fully automated diagnostic system based on breast thermography for early breast cancer identification. The main contribution of the proposed

framework is to extract the main breast mass information and then feed it to the ensemble classifier extreme gradient boosting (XGBoost) to classify breast tissue into normal or malignant classes diagnostically. To achieve this, we utilized the deep, pre-trained holistically-nested edge detector (HED) presented by Xie and Tu [10] to extract the characteristic features of the breast tissue. The HED has proven to be highly effective in identifying edges and contours in medical images, making it an ideal choice for our classification task. However, we found that integrating the Canny edge detector [11] can improve, to some extent, the classifier's accuracy. By combining these two edge detection methods, we were able to identify subtle differences in the texture and shape of breast tissue that may not have been detectable using only one method. This resulted in a more robust and accurate classification model that can aid in the early detection and treatment of breast cancer. Overall, our approach demonstrates the potential of combining advanced image processing techniques with machine learning algorithms to improve medical diagnosis and treatment outcomes.

The remaining sections of this paper are organized as follows: Section II provides the detailed methodology of data collection, feature extraction, and classification technique that have been followed in this study. Section III performs the results achieved and performance evaluation. In Section IV, the conclusion and future work are addressed.

II. METHODOLOGY

A. Data Collection

In this study, the breast thermography dataset employed was obtained from the DMR-IR database, which had been approved by the Research Ethical Committee of the Brazilian Ministry of Health (Approval No. CAAE: 01042812.0.0000.5243) [12]. The dataset consisted of breast thermal images acquired from a total of 296 patients, using a FLIR SC-620 camera with a resolution of 480×640 pixels. For each patient, the image collection process was conducted by a trained team under consistent room temperature conditions. Two types of images were captured: static and dynamic. In the static scenario, the patient was required to rest for a duration of 10 minutes, during which five thermal images were taken from different viewpoints. In the dynamic scenario, the patient's breast areas were subjected to cooling using an electric fan for approximately 5 minutes. Following the cooling phase, a sequence of 20 thermal images was captured, with each image taken at an interval of 15 seconds. This dynamic image collection process aimed to re-establish the patient's body's thermal equilibrium with the surrounding environment. For the purposes of this study, the static protocol was utilized due to its advantages of being faster, more comfortable for patients, and less prone to noise interference. To ensure data balance in the analysis, a total of 40 patients with confirmed breast abnormalities were included, utilizing the entire available data for this subgroup. Additionally, 40 individuals were randomly selected from the normal dataset to represent the control group in the analysis.



Fig. 1. Canny edge detection. (a) original thermal image, (b) the result of the edge detector, and (c) the result after applying weighted attributes and dilatation operation

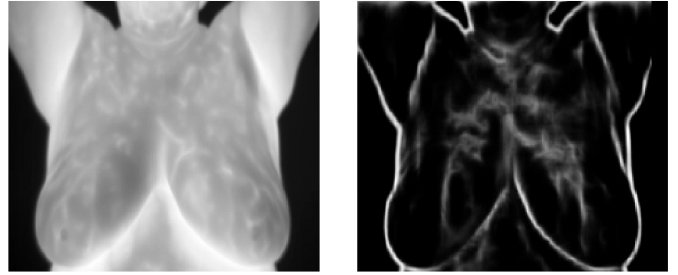


Fig. 2. Holistically-nested edge detection. (a) original thermal image and (b) the result of the edge detector

B. Feature extraction

In this study, edge detection played a crucial role in extracting valuable information from the breast images while reducing the amount of data that needed to be processed [13]. To achieve this, two edge detection techniques, namely the Canny edge detector and the holistically-nested edge detector, were employed to extract the characteristic features of the breast. The Canny edge detector was chosen due to its widespread usage and popularity in accurately identifying edges without being affected by noise distractions. This detector overcomes the limitation of not identifying weak edge points that are not connected to strong ones in an 8-connected manner. The Canny edge detection process involves multiple steps. First, the image is smoothed using a Gaussian filter to reduce noise interference. Then, local gradients are computed to determine the magnitude and direction of the intensity changes in the image. Pixels exhibiting local maxima in the direction of the gradient are then traced to identify the potential edges. Finally, a double thresholding technique is applied to classify the identified edges into strong and weak categories [14]. This ensures that both strong and weak edges are correctly preserved. To enhance the edge information obtained from the detection process, weighted attributes were applied to the retrieved edges. This step aimed to further highlight the important edge features within the image. Following this, a morphological dilatation operation was performed using a line-shaped structuring element. This operation aids in expanding and thickening the extracted edges, leading to a more robust representation of the breast's edge characteristics. As depicted in Fig. 1, an example breast thermal image is shown alongside its corresponding edges extracted using the Canny edge detector. The visual representation highlights the effectiveness of the edge detection technique in capturing the

prominent features of the breast image.

The effectiveness of edge identification algorithms has considerably improved with the utilization of deep convolutional neural networks (CNN) [15]. The HED is an end-to-end edge and boundary detection approach based on CNN in conjunction with deeply supervised nets that performs better than all other CNN. It has been effectively used in medical image segmentation [16]. A sample result for this detector is shown in Fig. 2.

C. Extreme Gradient Boosting

The extreme gradient boosting algorithm, originally developed by Chen and Guestrin [17], is a powerful and efficient end-to-end tree-boosting machine learning system. It operates on the principle of gradient-boosted ensemble learning, which allows for the creation of a highly accurate predictive model. One of the key advantages of XGBoost is its ability to minimize the loss function effectively, which leads to superior predictions compared to many other algorithms. By optimizing the loss function, XGBoost aims to find the best possible model fit for the given data. Moreover, the algorithm incorporates objective functions and regularization methods to mitigate the risk of overfitting, where the model becomes too closely tailored to the training data and fails to generalize well to new data points. In addition to its robustness in handling loss function optimization and preventing overfitting, XGBoost offers the capability to address common challenges encountered in machine learning tasks, such as missing values and unbalanced data. Missing values in the dataset can be handled seamlessly by XGBoost, ensuring that valuable information is still utilized in the modeling process. Similarly, the algorithm provides mechanisms to effectively handle imbalanced datasets, which often occur when one class is significantly more prevalent than the others. XGBoost employs techniques to adjust the learning process and allocate more weight to minority classes, thus improving the overall performance of the model. Given its remarkable capabilities and versatility, XGBoost has been chosen as the algorithm of choice in this study for the classification task. The algorithm's ability to optimize loss functions, produce accurate predictions, prevent overfitting, and handle common machine-learning challenges makes it a well-suited candidate for the task at hand.

III. RESULTS AND DISCUSSION

In this work, we proposed a fully automated system for early breast cancer detection from thermal images. The Holistically-nested edge detector was applied to the images to extract the characteristic features of the breast tissue. Besides, the canny edge detector was utilized to compensate for the missed edges and hence enrich the extracted breast information. The results of applying the two edge detectors are shown in Fig. 3. It is evident that the HED results in the multi-level extraction of breast thermal characteristics. On the contrary, the edges that the Canny detector recovered

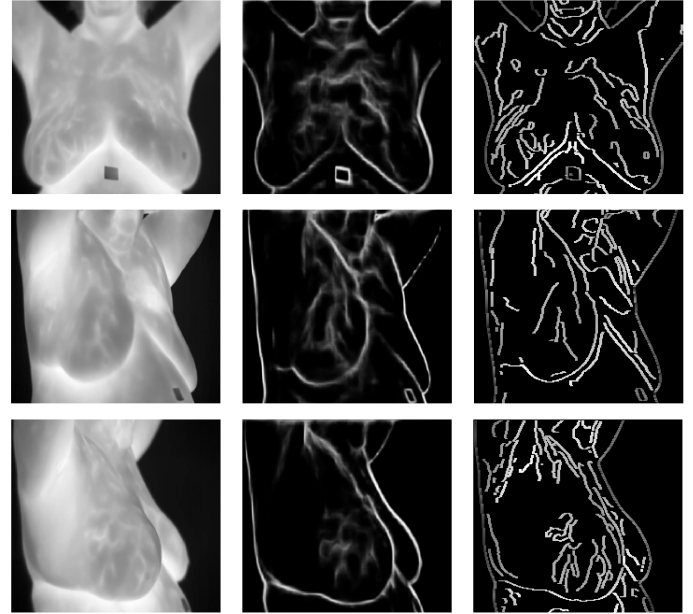


Fig. 3. Examples of breast information extraction. (a) original thermal image, (b) the result of the Holistically-nested edge detector, and (c) the result of the Canny edge detector

are inconsistent and unconnected. However, the Canny edge detector highlights several minor information gaps.

The characteristic information after being covered from the breast thermal images using the canny edge detector alone, the HED alone, and both were then fed to the XGBoost classifier. The classification model performance was evaluated by multiple measures, including accuracy, precision, recall, F1 score, and area under the receiver operating characteristic curve (AUC-ROC). The test results from 16 patients are presented in Table I. The classification performance by means of HED only achieved an accuracy of 96.1%, with a precision of 93% and recall of 100% and F1 of 96% as shown in Fig. 4. By Integrating the Canny edge detector, the classifier performance was improved to some extent, whereas the average values for the accuracy, precision, and F1 score increased by 1.3%, 2%, and 1%.

cross-validation results with a score of 94% showed that the suggested model is stable and has good generalizability to new and unexplored data. The AUC-ROC value of 99% (Fig. 5) demonstrates the system's capability to distinguish between normal and malignant individuals. To sum up, the HED with XGBoost classifier exhibits promising outcomes for thermography-based early breast cancer detection.

TABLE I
SUMMARY OF CLASSIFICATION PERFORMANCE RESULTS

Characteristic features	Accuracy	Precision	Recall	F1 Score
Canny	93.5%	92%	95%	93%
HED	96.1%	93%	100%	96%
HED & Canny	97.4%	95%	100%	97%

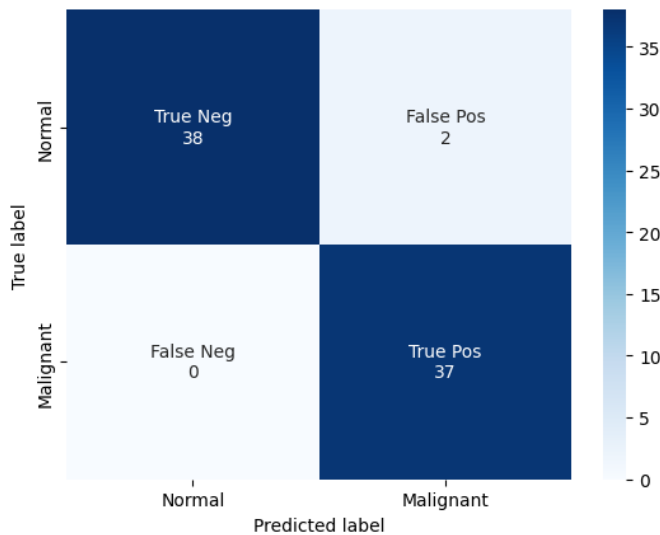


Fig. 4. Classification model confusion Matrix on the testing dataset

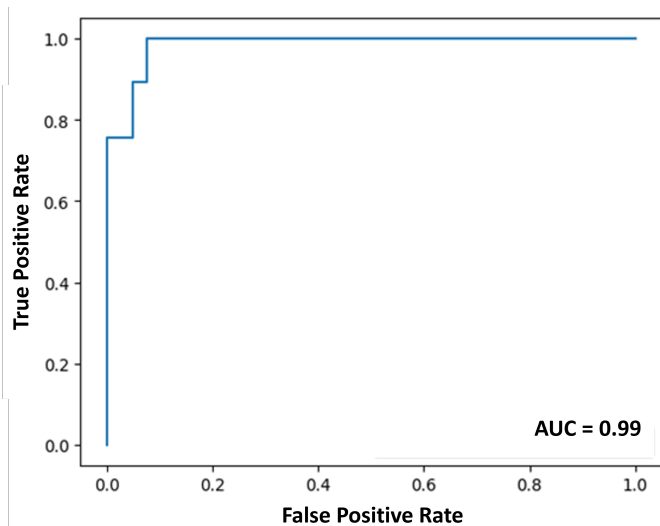


Fig. 5. AUC-ROC Curve from classifier evaluation on the training & testing

IV. CONCLUSION

In this study, we developed a new fully automated classification system for early breast cancer detection using thermography using the machine learning algorithm extreme gradient boosting. The system integrated the Canny edge detector with the deep pre-trained holistically-nested edge detector to extract the characteristic features of the breast tissue to train the classification model. Cross-validation was performed to evaluate the model's performance. The developed system achieved high accuracy of 97.4%, with precision, recall, F1 score, and AUC-ROC of 95%, 100%, 97%, and 99%, respectively. Due to its high accuracy and performance, the suggested model is a promising tool for thermography-based early breast cancer detection. This system could improve the efficiency and effectiveness of cancer diagnosis, leading to better patient outcomes.

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